```
function vectorScalingFactor(A, B, C, D)
                  #1
  v = (B.x-A.x, B.y-A-y)
  w = (D.x-C.x, D.y-C-y)
  u = (-v.y, v.x)
  return ((B.x-D.x) * u.x + (B.y-D.y) * u.y) / (w.x * u.x +
w.y * u.y)
```

#1 Method `vectorScalingFactor` takes four points, assuming they are the endpoints of segments `AB` and `CD`, and returns the scaling factor that need to be applied to vector `CD` in order to end exactly on segment `AB`.

When either value is 0 or 1, it means that `P` coincides with one of the segment's endpoints; if both `h` and `g` are equal to either 0 or 1, then we have an edge case where the two segments have a vertex in common. These two edge cases are shown in figure 15.20.

In conclusion, if we assume that all the 4 vertices are distinct, the two segments intersect if and only if both $0 \leq h \leq 1$ and $0 \leq g \leq 1$ (but only at most one of them is either 0 or 1).